Social Search on r/advice

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DATA CREATION

We extracted the most popular posts from the **r/advice** subreddit, selecting only those with at least 16 comments.
• For each post, we chose 16 relevant comments with at least 10 and no more than 150 upvotes, also including metadata.

WILSON SCORE

$$Wilson_{score} = \frac{\hat{p} + \frac{z^2}{2n} - z \cdot \sqrt{\frac{\hat{p}(1-\hat{p}) + \frac{z^3}{4n}}{n}}}{1 + \frac{z^2}{n}}$$

• $\hat{p} = \frac{x}{n}$ is the proportion of successes (upvotes / total_votes)
• x number of upvotes
• n total number of votes
• z z-value for the confidence level (e.g. 1.96 for 95%)

DATASET *POSTS*
- We collected posts that contained at least 16 comments.
- Each post is represented by the columns: **post_id**, **post_text** (title of the post), and **post_description** (full text of the post).

DATASET *COMMENTS*
- We extracted 16 relevant first-order comments with at least 10 and no more than 150 upvotes.
- Each comment is described by the columns: **comment_id**, **text**, **upvotes**, **replies**, **timestamp**, **karma_post** and **karma_comments**.

DATASET *QRELS*
- We ranked comments using the **Wilson score** to balance high upvote ratios with vote count.

- We assign a relevance level (1–4) to each comment using its Wilson score within the same post_id group.
  This is done by computing quantiles (Q1, Q2, Q3) for each group and assigning levels.

- To ensure a balanced and representative set of relevance judgments, we select a fixed number of top comments per relevance level (1–4) within each post, following number of upvotes.

STRUCTURE OF NOTEBOOK

REDDIT DATASET CONSTRUCTION
- Creation of *posts* and *comments* dataset
- Loading *posts* and *comments* dataset
- Creation of *qrels* dataset

EXPLORATIVE DATA ANALYSIS
- Distribution of posts length (in words) for *Title* and *Description*
- Distribution of comments lengths (in words)
- Top word frequency in posts (excluding stop words) for *Title* and *Description*
- Top word frequency in comments (excluding stop words)
- Correlation Matrix: Comment Features and Relevance Level

RETRIEVAL MODELS: BM25 and NEURAL RERANKING
- Baseline Retrieval (BM25)
- Neural reranking

FUNCTIONS for EVALUATION METRICS
- Precision and Recall
- MAP
- nDCG

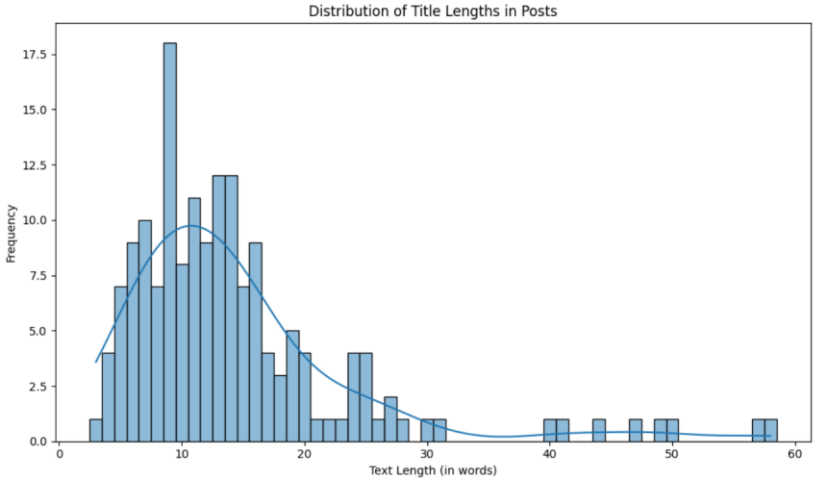
EVALUATION METRICS
BM25
- Precision and recall, MAP and nDCG

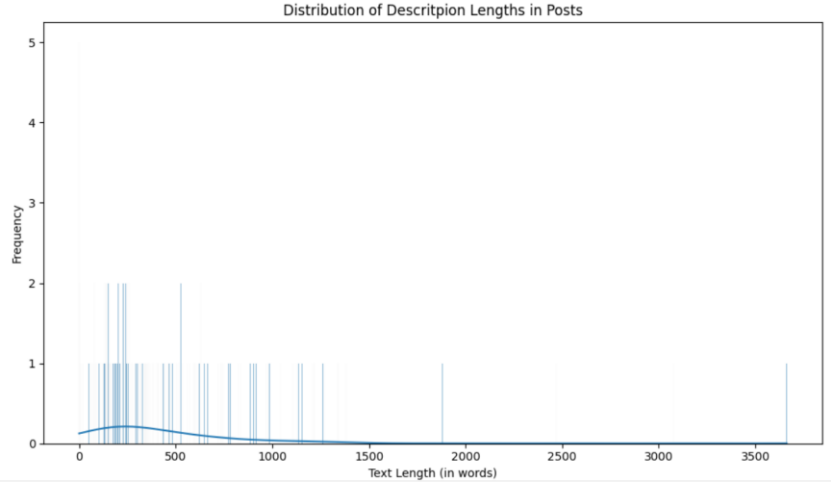
NEURAL RERANKING
- Precision and recall, MAP and nDCG

EXPLORATIVE DATA ANALYSIS - 1

The EDA is focus on:

- Distribution of **posts** length (in words) for *Title*
- Distribution of **posts** length (in words) for *Description*

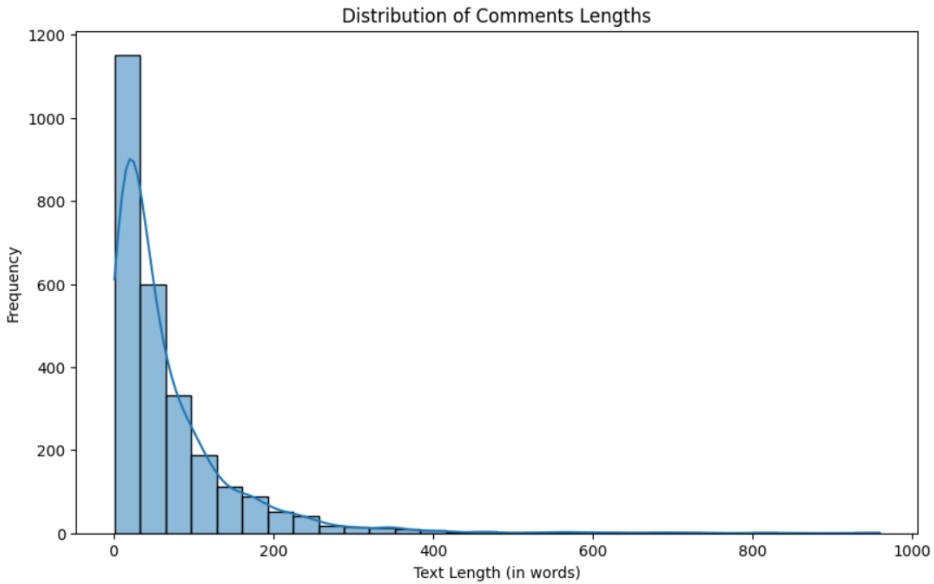




EXPLORATIVE DATA ANALYSIS - 2

The EDA is focus on:

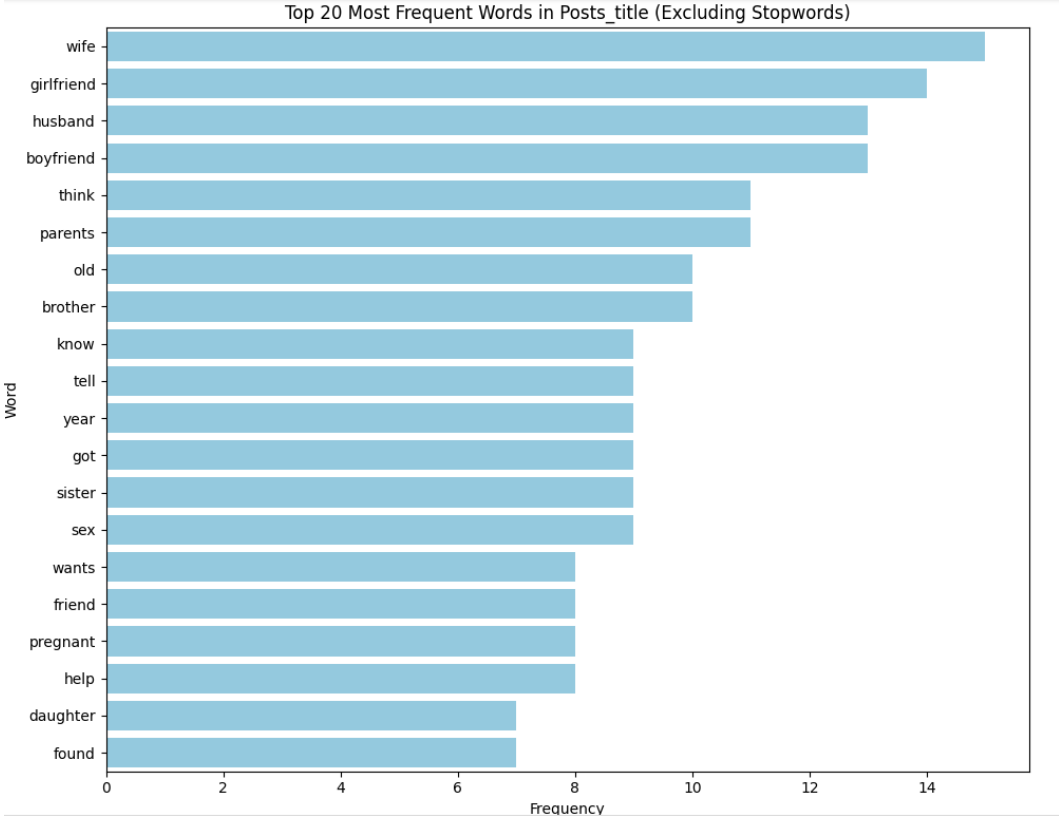
- Distribution of **comments** lengths (in words)



EXPLORATIVE DATA ANALYSIS - 3

The EDA is focus on:

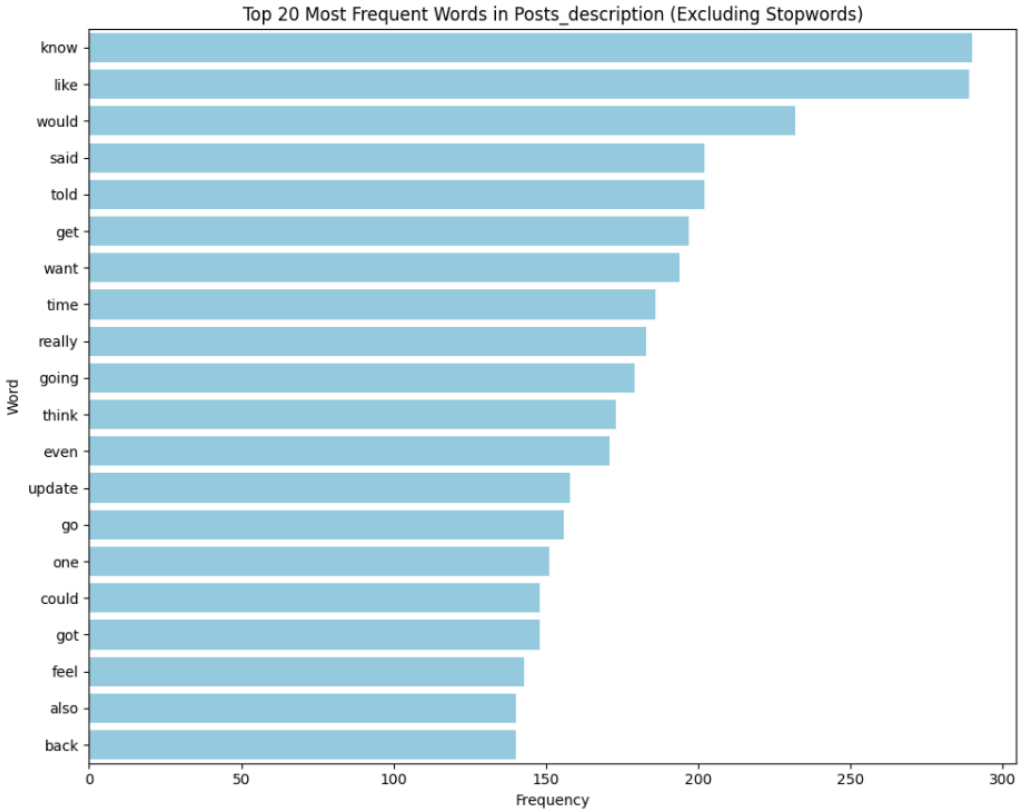
- Top word frequency in **posts** (excluding stop words) for *Title*



EXPLORATIVE DATA ANALYSIS - 4

The EDA is focus on:

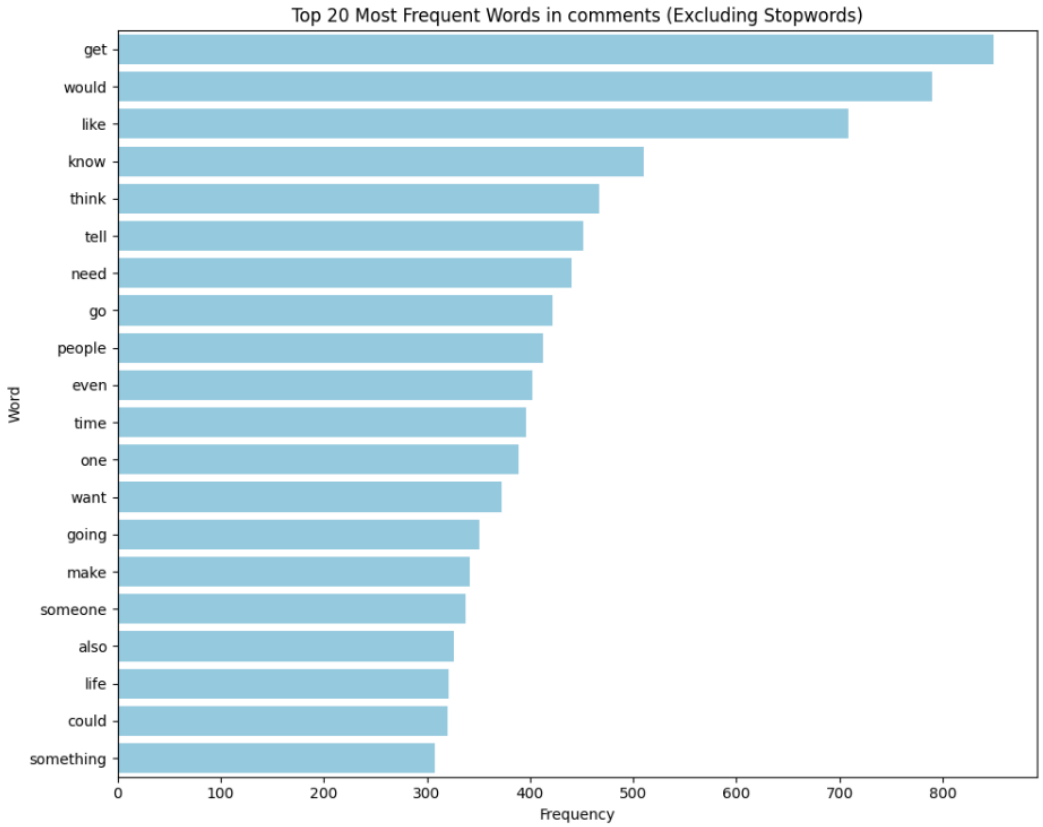
- Top word frequency in **posts** (excluding stop words) for *Description*



EXPLORATIVE DATA ANALYSIS - 5

The EDA is focus on:

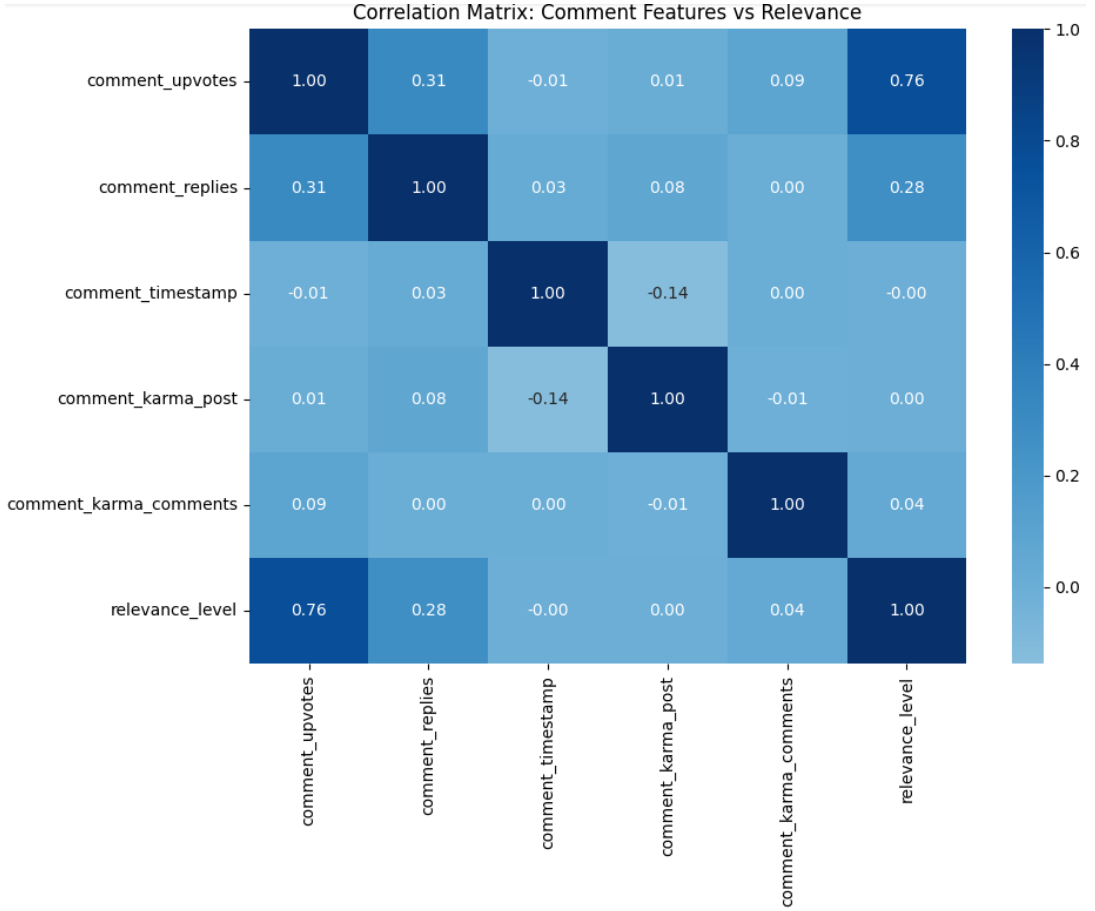
- Top word frequency in **comments** (excluding stop words)



EXPLORATIVE DATA ANALYSIS - 6

The EDA is focus on:

- Correlation Matrix: Comment Features and Relevance Level



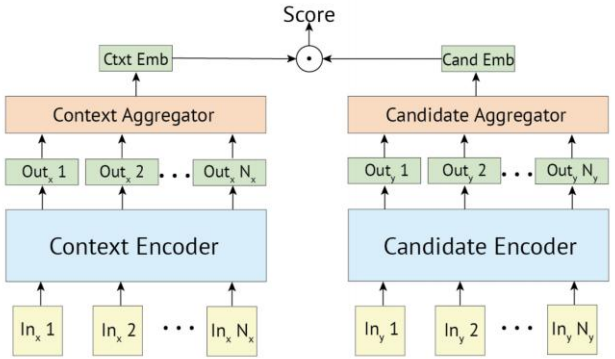
BM25 and NEURAL RERANKING

Best Matching 25 (BM25)

BM25 is a ranking algorithm used to measure how relevant a document is to a given query.
It belongs to the probabilistic model family and considers term frequency and document length.
Common terms are penalized to avoid bias toward frequent but uninformative words.
Its balance of simplicity and effectiveness makes it a strong baseline in search systems.

NEURAL RERANKING (Bi-Encoder)

Bi-Encoder neural reranking models encode the query and document separately into vectors.
Similarity is measured (e.g., via cosine similarity) to assess relevance between the two.
Since document embeddings are independent of the query, they can be precomputed and indexed.
This approach enables fast and scalable retrieval, ideal for large-scale search systems.



(a) Bi-encoder

EVALUATION METRICS

PRECISION
Measures the proportion of retrieved documents that are actually relevant. It evaluates how accurate the system is in returning only relevant results.

RECALL
Measures the proportion of relevant documents that have been successfully retrieved. It reflects the system's ability to find all relevant results.

Mean Average Precision (MAP)
Represents the average of the precision values at the ranks where relevant documents are found, averaged across all queries. It considers both relevance and ranking.

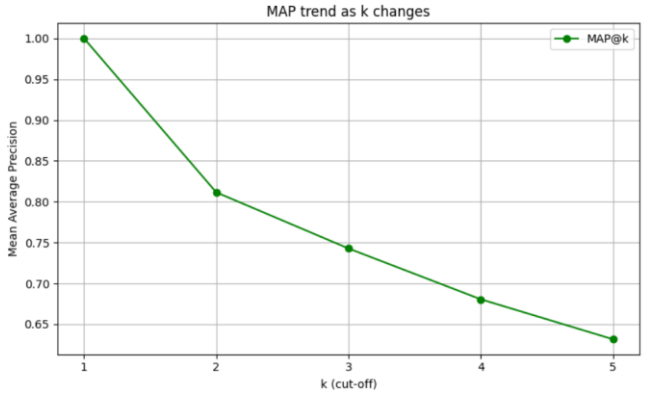
Normalized Discounted Cumulative Gain (nDCG)
Assesses the quality of the ranking by giving higher importance to relevant documents appearing at the top. It is normalized to allow fair comparison across different queries.

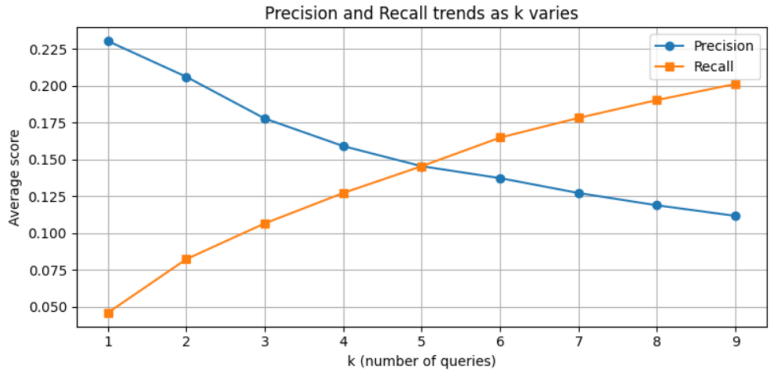
EVALUATION METRICS – BM25

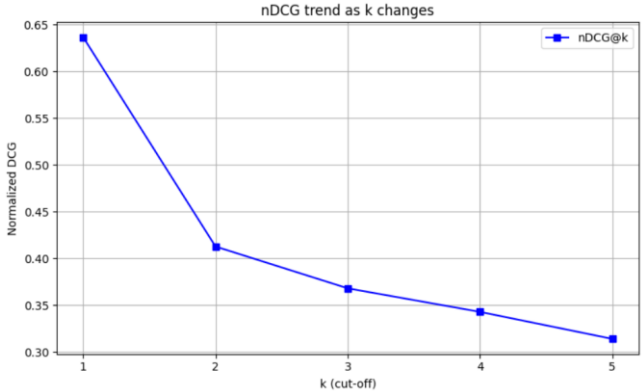
RESULTS

BM25 (@5)

PRECISION	RECALL	MAP	nDCG
0.15	0.15	0.63	0.31







EVALUATION METRICS – NEURAL RERANKING

RESULTS

NEURAL RERANKING (@5)

PRECISION	RECALL	MAP	nDCG
0.18	0.18	0.62	0.36

